

LAB 10 EXERCISE – Advance Data Analysis

Duration: 2 hours

Mode: Guided Lab and Hands on Practice

Module: CT127-3-2 Programming for Data Analysis

Lecturer: Dr. Kulothunkan Palasundram (Dr. Kulo)

Objective

By the end of this lab, you will have sufficient skills to write R scripts to perform advance data analysis to find answers for your analysis questions.

Dataset: SuperMarket_Cleaned.csv (Moodle).

Refer to Lecture 9 notes to perform this Lab 10 exercise.

Write R code to provide answers to these questions:

Part A – Correlation Analysis

1. Does increasing unit prices tend to increase sales?
2. Does quantity sold and unit price have a strong correlation with each other?
3. Is there a strong correlation between quantity and sales amount?

Part B – Comparative Analysis

4. Does payment method (Cash, Credit Card, Ewallet) show any pattern with sales amount?
5. Does branch location influence the correlation between price and sales?
6. Do males prefer Ewallet more than females?
7. Does customer type (Member vs Normal) influence payment method choice?

Part C – New Variables/Columns

8. Do high-rating transactions tend to use Ewallet more than Cash?
9. Are high-value purchases (sales > 500) more likely to be paid for by Credit Card?